

Experiment No. 3 – Viva Questions with Answers

Question 1. What is the difference between descriptive statistics and inferential statistics?

Answer:

Descriptive statistics summarize and describe the characteristics of a dataset, while inferential statistics use sample data to make predictions or draw conclusions about a larger population.

Descriptive Statistics

Summarizes data

Uses mean, median, mode, etc.

Does not make predictions

Inferential Statistics

Draws conclusions about a population

Uses hypothesis testing and confidence intervals

Makes predictions and decisions

Question 2. Explain the concepts of the Null Hypothesis (H_0) and Alternative Hypothesis (H_1).

Answer:

- Null Hypothesis (H_0): It assumes that there is no significant difference or relationship between the variables being studied.
- Alternative Hypothesis (H_1): It states that there is a significant difference or relationship between the variables.

Example:

- H_0 : Monthly income is the same for employees who leave and those who stay.
- H_1 : Monthly income is different for employees who leave and those who stay.

Question 3. What is a p-value? How is it used to make statistical decisions?

Answer:

A p-value is the probability of obtaining the observed result if the null hypothesis is true. It helps determine whether the result is statistically significant.

Decision Rule:

- If $p\text{-value} < 0.05$, reject the null hypothesis.
- If $p\text{-value} \geq 0.05$, fail to reject the null hypothesis.

Question 4. Differentiate between a t-test and ANOVA. In which situations is each test applied?

Answer:

A t-test compares the means of two groups, whereas ANOVA (Analysis of Variance) compares the means of three or more groups.

t-test

Compares two groups

One comparison

Example: Salary of Male vs Female employees

ANOVA

Compares three or more groups

Multiple group comparison

Example: Salary across different Job Roles

Question 5. What does the Pearson correlation coefficient indicate? What are its possible values?

Answer:

The Pearson correlation coefficient (r) measures the strength and direction of a linear relationship between two numerical variables.

Possible values:

- $+1$: Perfect positive correlation
- 0 : No linear correlation
- -1 : Perfect negative correlation

The value of r always lies between -1 and $+1$.

Question 6. Explain the significance of R^2 (Coefficient of Determination) in Linear Regression.

Answer:

R^2 (Coefficient of Determination) indicates how well the independent variable explains the variation in the dependent variable.

- $R^2 = 1$: Perfect prediction.
- $R^2 = 0$: The model explains none of the variation.

For example, an R^2 of 0.80 means that 80% of the variation in the dependent variable is explained by the regression model.

Question 7. Why is statistical analysis important before applying machine learning algorithms?

Answer:

Statistical analysis helps in understanding and preparing data before training a machine learning model. It helps to:

- Understand data distribution.
- Detect outliers.
- Identify relationships between variables.
- Validate assumptions.

- Select important features.
- Improve model accuracy and reliability.

Question 8. What assumptions should be satisfied before performing a t-test or ANOVA?

Answer:

Before performing a t-test or ANOVA, the following assumptions should be satisfied:

1. The observations should be independent.
2. The data should be approximately normally distributed.
3. The groups should have equal or similar variances (homogeneity of variance).
4. The sample should be randomly selected.

Question 9. How can regression analysis help organizations in forecasting and decision-making?

Answer:

Regression analysis helps organizations predict future outcomes based on historical data. It can be used to:

- Forecast sales and revenue.
- Predict employee salaries.
- Estimate customer demand.
- Analyze business trends.
- Support strategic planning and decision-making.

Question 10. Give two real-world applications where hypothesis testing is commonly used in data analytics.

Answer:

Two common real-world applications of hypothesis testing are:

1. Healthcare: Testing whether a new medicine is more effective than an existing treatment.
2. Marketing (A/B Testing): Comparing two advertisements or website designs to determine which performs better in terms of customer engagement or conversions.